Marupatla Jithin

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.88
Sri Chaitanya junior college	Class XII (TSBIE)	2020	93.00%
Kendriya Vidyalaya warangal	Class X (CBSE)	2018	76.00%

EXPERIENCE

Network Security Intern | STQC (ETDC), ECIL, Hyderabad

Ministry of Electronics and Information Technology (MeitY)

- **Led a team of 4 interns** to analyze website security reports, coordinate vulnerability reviews, and document key findings through a structured process that improved team-based security assessment and reporting.
- **Analyzed High, Medium, and Low-risk vulnerabilities** by reviewing **VAPT reports**, identifying affected systems, understanding security issues, and mapping each finding with suitable remediation steps and recommendations.
- **Improved security reporting quality** by organizing vulnerability details, affected versions, risk levels, and remediation steps into structured reports while gaining practical exposure to **GIGW testing requirements**.

THESIS

Adaptive Inference Routing (AIR) -- Learning-Based Threshold Routing & Communication-Efficient Offloading for Edge-Cloud Perception (JNCA Journal Extension)

- **Reducing communication overhead** in AIR by implementing **adaptive resolution and compression** before offloading, lowering transmitted data and inference latency while preserving perception **accuracy**.
- **Eliminating manual threshold recalibration** across deployments by developing **learning-based routing** for AIR, replacing static bandwidth/resource rules with an **online regret-bound algorithm** that adapts to real-time **conditions**.
- **Strengthening the technical foundation** by surveying 15+ papers on **contextual bandits, cost-function learning, hierarchical inference**, benchmarking candidates against AIR's baseline for deployment **decisions**.

PROJECTS

NAVINUX -- GenAI Linux Operations Assistant

- Architected an **AI-powered Linux assistant** translating natural language into safe, executable diagnostics.
- Engineered a registry-based trust layer isolating LLM intent from execution, enforcing four invariants across 55 tools
- Achieved 100% injection resistance across 5,201 attacks and **93.3% retrieval accuracy** via build-gated evaluation.

AI Travel Assistant with Memory & Semantic Caching

- Engineered an **AI travel assistant** using FastAPI, LangGraph, Redis, and Gemini.
- Implemented **semantic caching** via SHA-256 fingerprinting, cutting redundant LLM calls significantly.
- Orchestrated the pipeline with LangGraph, optimizing latency across FastAPI REST endpoints.

Hidden Gods of Bharath -- AI Cultural Heritage Platform

- Developed an interactive India map interface visualizing archaeological and cultural data for regional village deities.
- Integrated Gemini API to build a conversational **AI chatbot** answering user queries on deity origin stories.
- Implemented multilingual translation using Text to Speech and Bharat GPT for regional language accessibility.
- **Added accessibility features** including read aloud narration and adjustable font size for visually impaired users.

POSITIONS OF RESPONSIBILITY

Assistant Manager, E-Cell (2025): Led a 4-member team to organize E-Summit, coordinating logistics and execution

end-to-end.

ACHIEVEMENTS

- **CDAC National-Level Hackathon (2026):** Reached the **Grand Finale** nationwide among 9,000+ participants.
- **RISHI.AI Hackathon (2025):** Secured **2nd place** among college-level teams for *Hidden Gods of Bharath*.
- Published a peer-reviewed **research paper** in **IJRASET**, an international journal, detailing the project's design and outcomes

SKILLS

Languages: Python,C++,HTML,CSS

AI & ML: Machine Learning, Deep Learning, NLP, GenAI, PyTorch, RAG Pipelines, Vector Databases, LLM Fine-Tuning

NLP & LLMs: LangChain, Intent Extraction, Slot Extraction, LLM Agents, Multi-Agent Systems.

Backend & Databases: FastAPI, Redis, MySQL

EXTRACURRICULAR

Sports: Competed in Swimming, Squash, and Tennis at Milan (intra-college) and Diesta (inter-department) tournaments; also play Table Tennis and Cricket (specializing in bowling).

Film Making: Write original short stories and produce short films, combining scriptwriting with hands-on filmmaking and visual storytelling.